

Carlos Hernández

Senior AI Platform Engineer

Observable, version-controlled, costed, deletable LLM systems.

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SUMMARY

Platform engineer with 8+ years building production cloud systems and the engineering organizations behind them, now building the platforms that run LLM systems in production — multi-tenant RAG, agent runtimes, evaluation harnesses, and the delivery infrastructure underneath them. **Two tracks by design:** contract platform engineering as a stable base, and end-to-end ownership of the AI platform at a company I helped build from formation — now consolidating both into one full-time Senior/Staff AI Platform role. Track record shipping LLM products 0→1 in enterprise contexts, with 20%+ cloud cost reduction and 95%→99.5% availability uplift. Currently completing a Master's thesis on comparative NLP at Universitat de Barcelona.

CORE TECHNICAL SKILLS

AI / LLM Engineering · Production RAG architectures · Hybrid retrieval & Reciprocal Rank Fusion · Agent runtimes & tool calling · Multi-tenant isolation · Prompt engineering, versioning & progressive retry · LLM evaluation and adversarial red-teaming · LLM observability (Langfuse) · Vector search (pgvector, HNSW) · Google ADK · Gemini · Claude API

Platform Engineering & SRE · CI/CD at organizational scale (GitLab CI) · Reusable pipeline component libraries · Infrastructure as Code (Terraform, versioned modules) · Progressive delivery · DORA metrics · Site reliability & SLOs · FinOps and cost optimization · OpenTelemetry · Vault

Machine Learning (academic) · NLP text classification · Transformer fine-tuning (DistilBERT) · Classical ML baselines (TF-IDF + SVM, XGBoost) · Benchmark design and data-leakage detection · Statistical significance testing (McNemar) · ONNX deployment

Cloud & Infrastructure · GCP (Cloud Run, Cloud Run Jobs, Vertex AI, Cloud SQL, Pub/Sub, API Gateway) · AWS (multi-account, OIDC, Cloud Control API) · Kubernetes · Docker · Keycloak / JWT

Languages & Tooling · Python (async, FastAPI, Pydantic, SQLAlchemy) · TypeScript / Next.js · SQL / PostgreSQL · pytest

PROFESSIONAL EXPERIENCE

EsePlus Group · EsePlus & Alilo

Apr 2019 – Present

San Salvador, SV

Joined as Tech Lead at company formation and built the engineering organization from zero — 30+ hires across web, mobile, DevOps/SRE, QA, design and support — establishing the technical foundations and operational practices that scaled with the group.

Head of AI Platform Engineering

Oct 2025 – Present

- Architected and shipped **Alilo LXD AI** — an async generative-AI backend that turns uploaded documents into complete corporate microlearning experiences. Multi-stage pipeline coordinating document analysis, content generation and activity design, with backpressure handling and client-side streaming feedback.
- Architected and shipped **Alibot** — a multi-tenant RAG conversational backend: hybrid retrieval fusing dense HNSW vector search, full-text and fuzzy matching via Reciprocal Rank Fusion; Matryoshka embeddings; Google ADK agent runtime with multimodal tool calling. Both APIs designed agnostic and composable — invocable iteratively or as a sub-agent from higher-level orchestrators.
- Built a **defense-in-depth LLM safety stack**: input sanitization, output groundedness classification, structured refusal pools, and automated adversarial regression as a merge gate — zero jailbreaks at test time across the adversarial suite.
- Established **dual observability for AI workloads**: OpenTelemetry for infrastructure combined with Langfuse for AI quality (tokens, cost, prompt/response), unified via shared trace identifiers — enabling prompt iteration driven by production evals, not intuition.
- Built versioned prompt management with 3-level progressive retry and AI-assisted field sanitization, **cutting token cost ~90%** on validation failures vs. full-prompt retries; cut conversational context 70K→16K tokens through history truncation.
- Designed a custom rate limiter for Gemini API orchestration (concurrency control with exponential backoff), eliminating 429 errors at scale, and migrated heavy document processing from synchronous HTTP to an async architecture on Cloud Run Jobs.

Chief Technology Officer

Oct 2022 – Oct 2025

- Improved group availability from **95% → 99.5%** through SLO-driven strategy, standardized golden paths, and progressive delivery with canary deployments.
- Achieved **20%+ reduction in cloud spend** through FinOps practices, rightsizing and automated resource optimization.
- Standardized CI/CD pipelines and Infrastructure-as-Code patterns across services, eliminating manual deployment errors.

Tech Lead

Apr 2019 – Oct 2022

- Set the initial architecture, delivery practices and hiring bar for the engineering team as the company scaled from formation to a multi-product organization.

Stack: GCP · Vertex AI / Gemini · pgvector · Google ADK · FastAPI · GitLab CI · OpenTelemetry · Langfuse · Keycloak

Remote

- Designed and shipped a **shared CI/CD delivery platform** — a library of reusable components (install, lint, test, build, deploy) that any repository in the organization consumes in three lines of config, with three consumer repositories running on it in production, branch-routed deployment to per-team sandboxes, and DORA metrics instrumented end to end.
- Cut pipeline **wall clock 52%, runner time 54% and jobs per pipeline 62%** — measured p50 across a 14-day, 295-pipeline window rather than a single run. The largest single win was structural: the build never consumed lint's output, so the quality gate moved downstream to block publishing instead of compiling.
- Built a **self-service platform for AI SRE agents** that provisions a team-scoped incident-response agent — its integrations, permissions and behaviour — from a single block of configuration; onboarding a team is a merge request rather than a project.
- Made agent behaviour reviewable: playbooks and standing instructions ship as **versioned packages pinned per agent**, so changing how an agent behaves is a reviewed diff with an approver instead of an unauditable console edit. Every integration read-only by construction; the agent recommends, humans act.
- Implemented cross-account authentication via OIDC with role-hop and no standing credentials in CI, with least-privilege service identities scoped per integration.

Stack: GitLab CI · AWS (multi-account, OIDC, Cloud Control API) · Terraform · Vault · Datadog · Kubernetes

FullStack Labs · DevOps Engineer · Contract

Oct 2022 – Feb 2025

Remote

- Built and maintained GitLab CI/CD pipelines for Kubernetes workloads, reducing deployment time and eliminating manual deployment errors.
- Automated infrastructure provisioning with Terraform on AWS, enabling consistent environments across teams, and partnered with product teams to reduce release friction.

Defensoría del Consumidor · Systems & Database Administrator

Jul 2018 – Apr 2019

San Salvador, SV · National consumer-protection agency

Managed server infrastructure for a government agency, maintaining **99.9% uptime** with automated secure backup systems; led internal improvements aligned with ISO 9000.

SELECTED PROJECTS**Clarytia** · Multi-tenant RAG platform · designed, built and audited solo

Private beta

A verifiable internal-knowledge assistant: every answer carries its source document, page and confidence tier, and returns “I don't know” rather than improvising. Database-per-tenant isolation, so deleting a customer is a DROP rather than a query that has to cover every table. Five ranking lanes — dense HNSW, full-text, trigram, unaccent and fuzzy — fused by Reciprocal Rank Fusion in a single SQL query, with no external search service. Access control enforced inside the retrieval path, not as a post-filter. Knowledge-gap detection clusters unanswered questions into named gaps ranked by distinct users, with resolution gated on attaching a document and an async re-check that reopens the gap if the document still doesn't answer it.

2,814 tests across 232 files (>90% coverage) · 37 ADRs · 105 endpoints · generative red team gated at every deploy with zero leaks · 189 commits in 5 months, single author.

Master's Thesis · Comparative NLP for transaction classification

Defense Sep 2026

Comparative evaluation of classical baselines (TF-IDF + SVM, XGBoost), embedding-based neural networks and transformer fine-tuning (DistilBERT) for bank-transaction categorization, judged on accuracy, robustness to noisy text, statistical significance and total cost of ownership. **Found the public benchmark was leaking:** its datasets are template-generated, so a random split puts nearly every merchant in both train and test. Rebuilt the evaluation by degrading the test set with four realistic noise types — and wrote the prediction before running it, then reported being wrong. At 30% noise the linear model retained 89.9% of its score against DistilBERT's 84.2%, and McNemar moved from $p=1.00$ to $p=1.3 \times 10^{-78}$ in its favour. The transformer cost 356× the parameters and 15× the inference latency to tie on clean text and lose on dirty. The winning model exports whole to a 1.3 MB ONNX graph that runs in-browser: \$0 to serve, and the data never leaves the device.

Engineering writing · carloshdez.com

Weekly field notes on production LLM systems, published in English and Spanish: prompt versioning as schema migrations, what “observable” actually means for an LLM app, adversarial regression as a merge gate, and the thesis results above. Public talk: “Cursor + MCP — when your IDE knows your stack” (2026).

EDUCATION**Master's in Machine Learning & AI (60 ECTS)** · OBS Business School / Universitat de Barcelona, Spain

Oct 2025 – Oct 2026

In progress · Thesis on comparative NLP for text classification

Executive Master's in Artificial Intelligence · Instituto de Inteligencia Artificial, Alicante, Spain

Apr 2024

B.Sc. in Systems Engineering · University of El Salvador, El Salvador

Jun 2017